BrandGate — Intelligent Brand Governance Platform

The automated pre-publication gate every marketing post must pass before it reaches customers.

CORE GUARANTEE

“No citation, no flag.”

A finding with no valid vault citation ID must **NEVER** be shown. Every flag is strictly grounded in an active brand-vault record. If a promotional claim cannot be tied to an explicit rule or verified fact, zero violation is raised. Zero hallucinations, zero ungrounded rejections.

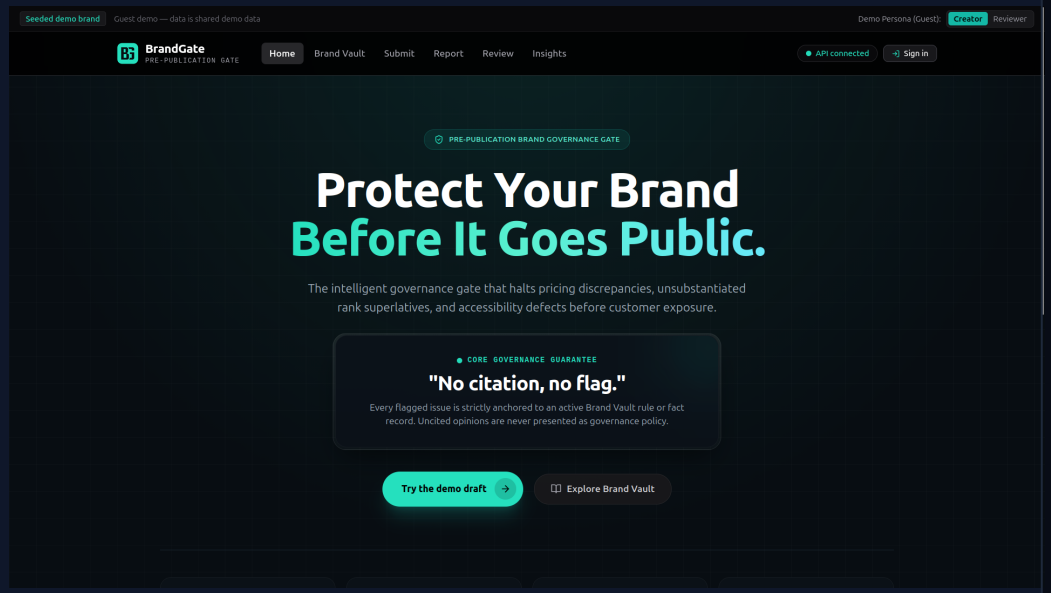
Dual-Persona Security & Velocity:
- **Creators:** Instant inline feedback & 1-click remediation without creative friction.
- **Reviewers:** Strict 403 author-approver separation, immutable audit logs, locked snapshots.

LIVE DEPLOYMENT & TEAM METADATA

- **Live App (Judge Link):** https://brand-gate-rho.vercel.app/
- **Backend API:** https://brandgate-codex-48.onrender.com (Render FastAPI)
- **GitHub Repo:** github.com/manthansingh26/BrandGate-Codex-48
- **Git Head:** 37 commits, current **059aedce** (Verified 2026-11-02)
- **Team:** SOLO team **STRAW HAT PIRATE** | Leader: **Manthan Shravansingh Rajpurohit**
- **Access:** Zero sign-in wall — complete evaluation accessible in Guest Demo mode.

Design Philosophy: Dark Space Navy theme (#05070E), Linear/Stripe Bento grid layout, WCAG AA contrast validation, zero-login guest evaluation.

LIVE PRODUCTION INTERFACE (ZERO SIGN-IN GUEST DEMO)



Live hero interface at brand-gate-rho.vercel.app — featuring seeded demo brand switcher, API health status, and 1-click demo draft kickoff.

Judge Path in 60 Seconds — Numbered Visual Workflow

Zero sign-in evaluation journey: from raw festive promotional copy to an immutable approved compliance snapshot.

STEP 1: KICK OFF

Try Demo Draft

On hero, click primary teal button. Seeds Brand A cookware draft: *"Diwali Dhamaka! India's No. 1 cookware brand... Rs. 999!"*

STEP 2: REVIEW

Deterministic Engine

Click **Review Draft**. Deterministic regex & vault engine executes with secondary Gemini advisory validation.

STEP 3: INSPECT

1x F-12 + 2x R-04

Flags Rs. 999 vs Rs. 1,299 (F-12 Contradiction) and 'No. 1' / 'clinically proven' (R-04 Evidence Needed). **0 uncited**.

STEP 4: APPLY FIX

1-Click Remediation

Click Apply Fix for each flag. Updates text cleanly to locked final text: *"...trusted cookware brand... Rs. 1,299!"*

STEP 5: PENDING

Pending Review State

Version 4 generated in **Pending Review**. System never prematurely displays 'Clear to Publish' before review.

STEP 6: REQUEST

Request Approval

Switch to Review tab. Creator requests approval. Status advances to pending_approval server-side.

STEP 7: TOGGLE

Reviewer Approval

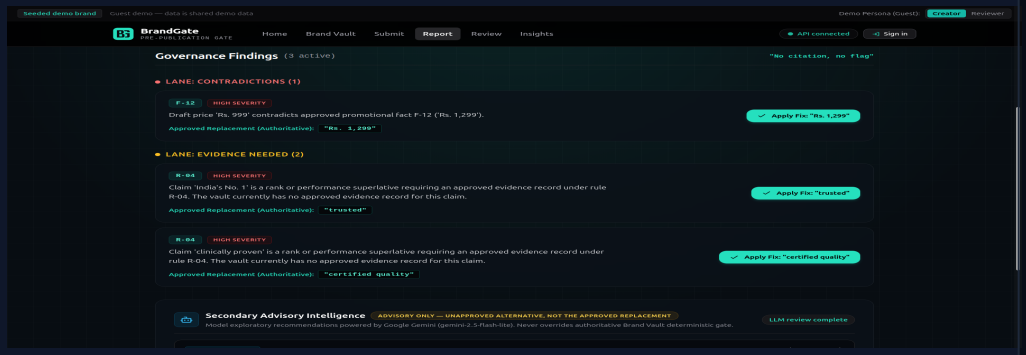
Toggle role to Reviewer in navbar. Click Approve Draft. Creator self-approval strictly blocked with **HTTP 403**.

STEP 8: LOCKED

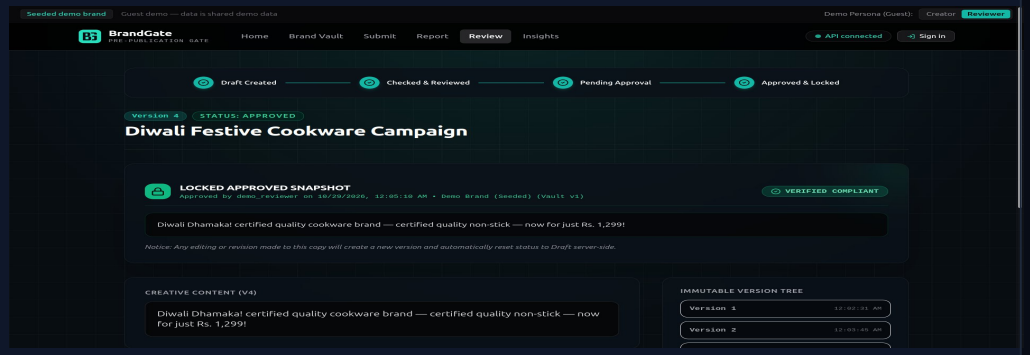
Approved Snapshot

LOCKED APPROVED SNAPSHOT card appears with **VERIFIED COMPLIANT**, approver role (demo_reviewer), and audit trail.

INITIAL GOVERNANCE REPORT (1x F-12 + 2x R-04)



LOCKED APPROVED SNAPSHOT (VERIFIED COMPLIANT)



Three Locked Round 3 Improvements (Built & Live-Verified)

All three improvements implemented, tested, and verified live on production in official required sequence.

1. MULTI-BRAND WORKSPACES

Tenant-Scoped Isolation Across Drafts & Vaults

• **Brand A (Demo Brand Cookware):** 12 Vault Items (F-12, A-02, R-04, R-09, T-01..T-04, P-01 #1E3A8A, P-02 #D97706, P-03 #F8FAFC, P-04 #0F172A).
• **Brand B (NovaTech Audio):** 5 Vault Items (F-20 Rs. 4,999, R-15, T-10 'world best', P-10 #06B6D4, P-11 #0B0F19).
• Zero cross-brand citation leakage: Brand B review (F-20, R-15) never cites Brand A cookware items.
• **How Judge Tests Live:** Click top-left workspace selector or homepage workspace pills to switch between Demo Brand and NovaTech Audio — inspect independent forms, vaults, and rules.

2. DOCX / PDF EXTRACTION

Document Governance with Page Badges & 10MB Limit

• **Order Preservation:** DOCX parser extracts paragraphs and tables in true document body order.
• **Multi-Page Page Badges:** PDF text extracted with exact page attribution — findings surface Page 1 vs Page 2 badges directly on finding cards.
• **Honest Scanned-PDF Limit:** Scanned PDFs without text layers explicitly report no OCR support — zero fake passes or synthetic hallucinations.
• **How Judge Tests Live:** On Submit tab, upload DOCX or multi-page PDF up to 10MB to see document text and Page N badges; test scanned PDF for honest limit notice.

3. INSIGHTS V2 & HARDENED IMAGES

PostgreSQL Aggregation & Robust Visual Governance

• **Grounded Insights v2:** Aggregated database metrics per brand with directional trend (IMPROVING/WORSENING) and fix-first directives. All counts are cumulative across history.
• **Approximate Palette Sampling:** Image color sampling explicitly labeled 'Approximate', generating advisory Human Decision findings (never Contradiction).
• **Reviewer-Only Logo (A-02):** Logo verification strictly requires reviewer confirmation — no fake CV.
• **How Judge Tests Live:** Open Insights tab to inspect live metrics; attach image on Submit to test Approximate palette sampling.

PLUS LIVE-VERIFIED RELIABILITY & LAYOUT HARDENING

• **Brand-Switch Restoration:** Never blank, never requires page reload — switching brands instantly restores the cached latest draft and current-version findings with zero blank renders.
• **Pending Review Interim:** Applying fixes creates new draft version strictly marked *'Pending Review'* — never prematurely labeled 'Clear to Publish' before review.
• **DOCX Reliability:** 0 *'Failed to fetch draft'* errors across consecutive reviews with document parsing resilience and retry guards.
• **Single-Row Homepage Pills & Viewport-Safe Navbar Dropdown:** Single-row layout on desktop (w-fit, flex-nowrap) with center-aligned mobile wrap; navbar brand dropdown permanently anchored to trigger left edge with viewport bounds clamp.

Complete Features List & Capabilities Matrix

End-to-end brand protection across advertising copy, document attachments, creative assets, and sign-off workflows.

BRAND VAULT TRUTH ENGINE

Centralized truth repository grouping approved facts (F-12, F-20), mandatory negative rules (R-04, R-15), approved terminology (T-01..T-04), and brand color palettes (P-01..P-11).

DETERMINISTIC REVIEW GATE

Authoritative primary server gate executing regex pricing reconciliation, superlative evidence checks, alt-text accessibility, and WCAG contrast validation.

GROUNDED GEMINI ADVISORY

Secondary compliance suggestions powered by Google Gemini (temperature 0.2), strictly bounded by retrieved vault items via keyword fallback and validated by citation validator.

ONE-CLICK REMEDIATION

Constructive fixes replacing prohibited superlatives with approved brand terminology (T-04 'trusted cookware brand', T-03 'certified quality') and approved pricing.

SERVER APPROVAL STATE MACHINE

Server-enforced status transitions: draft → checked → pending_approval → approved. Blocks creator self-approval (403), halts on open findings, resets to draft on edit.

DOCUMENT & MULTI-PAGE GOVERNANCE

Upload DOCX and multi-page PDFs up to 10MB. Findings surface exact Page N badges; unextractable assets flag 'Could not sample — human decision' with no fake pass.

Governance Dimension	Creator Experience	Reviewer / Compliance Role	System / Safety Guarantee
Promotional Pricing	Instant inline flag with 1-click update to F-12 (Rs. 1,299)	Verified against active approved promotion facts	`Deterministic regex pricing reconciliation`
Superlatives & Claims	Constructive replacements (e.g. 'trusted', 'certified quality')	Evidence-needed lane — superlatives held until clinical proof	`Rule R-04 / R-15 grounded claim engine`
Color Palette & Images	Palette check labeled 'Approximate'; alt-text reminder (R-09)	Reviewer confirmation required for logo placement (A-02)	`WCAG AA contrast & approximate palette math`
Sign-off & Publishing	Request Approval button; self-approval strictly prevented	Approve Draft button produces LOCKED APPROVED SNAPSHOT	`HTTP 403 enforcement & version immutability`

Technical Architecture & Implementation Summary

Dual-deployment monorepo: React 18 + Vite on Vercel CDN, FastAPI on Render, Supabase PostgreSQL with fail-closed safety.

SYSTEM ARCHITECTURE & EXECUTION TOPOLOGY

[CLIENT LAYER — VERCEL CDN] React 18 + TypeScript + Vite + Tailwind CSS | Zero-Login Guest Demo Mode | WCAG 2.1 AA Contrast Tools ■ ▼ HTTPS / REST (CORS Allowed: localhost, brand-gate-rho.vercel.app) **[API GATEWAY & BUSINESS LAYER — RENDER WEB SERVICE]** Python 3.11 + FastAPI + Uvicorn | 70s Health Timeout & 90s Retry Cadence ■ ■■■

Testing Verification & Real Database Insights

100% verified local automated test suites paired with live production PostgreSQL cumulative analytics.

LOCAL AUTOMATED TEST SUITES (100% PASSING)

• **Phase 4 Hardening** (tests/test_phase4_hardening.py): **10 / 10 passed** • **Brand-Switch Restoration** (tests/test_brand_switch_restoration.py): **6 / 6 passed** • **Phase 4.2 Hardening Suite** (tests/test_phase4_2_all.py): **5 / 5 passed** • **Document Governance Suite** (tests/test_document_governance.py): **8 / 8 passed** • **Homepage Workspace Selector** (tests/test_homepage_workspace_selector.tsx): **4 / 4 passed** • **Navbar Dropdown Clipping** (tests/test_navbar_dropdown_clipping.tsx): **3 / 3 passed** • **Frontend Production Build** (cd frontend && npm run build): **0 errors (1.98s)**

Live Production Verification Log:
• **2026-11-01 23:21 IST:** FULL JUDGE GO (PASS) — Full judge path verified live on Vercel.
• **2026-11-02 00:29 IST:** Dropdown Left Clipping (FAIL) — Observed live, reproduced & fixed.
• **2026-11-02 01:30 IST:** FULL JUDGE GO (PASS) — Post-fix end-to-end judge path verified live.
**Note: Mobile 360px and 120% zoom verified via automated layout geometry, not physical browser.*

POSTGRESQL DATABASE INSIGHTS (LATEST VERIFIED SET)

Cumulative Aggregates Across Full Evaluation History (Oct 2026 – Present)

Demo Brand (Seeded) [Captured: **2026-11-01 19:59:46 UTC**]: • Cumulative Drafts: **312** | Total Flags: **990** • Top Repeat Violation: **R-04** (571 occurrences) • Directional Trend: **IMPROVING** • Dispositions: **162** Clear to Publish | **33** Clear with Edits | **116** Hold for Legal **NovaTech Audio (Demo)** [Captured: **2026-11-01 19:56:15 UTC**]: • Cumulative Drafts: **69** | Total Flags: **212** • Top Repeat Violation: **R-15** (139 occurrences) • Directional Trend: **WORSENING** • Dispositions: **9** Clear to Publish | **1** Clear with Edits | **59** Hold for Legal

Complete Multi-Brand Isolation: Switching brand selector instantly updates metrics with zero cross-brand contamination. All numbers represent real database rows — zero synthetic or fabricated values.

LIVE POSTGRESQL GOVERNANCE INSIGHTS DASHBOARD



Product Roadmap & Future Plan (Post-Codex Proposals)

Strategic architecture enhancements proposed for enterprise rollout beyond the official hackathon release.

PROPOSAL 1: OCR ENGINE FOR SCANNED DOCUMENTS

Tesseract / AWS Textract Pipeline for Scanned Media

- Ingest scanned PDF print advertisements, flyers, and photographed packaging labels.
- Multi-engine text extraction with confidence scoring to prevent OCR noise from triggering false compliance flags.
- Bounding-box visualization highlighting the exact physical location of pricing or claim violations on the print proof.

PROPOSAL 2: FIGMA & DESIGN TOKEN SYNC

Automated Vault Synchronization with Brand Systems

- Direct bi-directional integration with Figma Design Tokens and Notion/Confluence brand guidelines.
- Rule updates in the brand book automatically propagate into BrandGate Vault facts, rules, and palettes.
- Webhook notifications alerting compliance reviewers whenever a marketing team updates promotional pricing.

PROPOSAL 3: HEADLESS GOVERNANCE API & CI/CD GATE

Pre-Commit Compliance Checks for Marketing Repositories

- GitHub Actions & GitLab CI integrations to validate CMS copy, markdown articles, and banner assets in PRs.
- Figma Plugin providing real-time inline compliance feedback directly within designer canvas.
- Programmatic approval token webhooks triggering automated social media publishing upon snapshot sign-off.

PROPOSAL 4: ENTERPRISE MULTI-TENANT SSO & RBAC

SAML 2.0 / Okta Integration & Organizational Tenancy

- Enterprise single sign-on (SSO) with Okta, Azure AD, and Google Workspace.
- Granular role-based permissions: Creator, Copy Editor, Legal Reviewer, Agency Contributor, Compliance Officer.
- Complete tenant data isolation with customer-managed encryption keys (CMEK) and audit export compliance.

Honest Limitations & System Disclosures

Transparent disclosure of operational boundaries, trade-offs, and verification scope across the platform.

1. SCANNED PDF INGESTION (NO OCR)

Evaluates digital text layers only. Scanned image PDFs without embedded text layers cannot be evaluated for text; Optical Character Recognition (OCR) is not performed and synthetic text is never fabricated.

2. APPROXIMATE PALETTE SAMPLING

Image color sampling math is explicitly labeled 'Approximate'; rule P-01 produces advisory Human Decision even on on-palette images to prevent automated false rejections.

3. REVIEWER-CONFIRMED LOGO (A-02)

Automated computer-vision logo recognition is never claimed. Rule A-02 strictly requires manual reviewer confirmation to prevent deceptive logo approvals.

4. DETERMINISTIC GATE AUTHORITY

Google Gemini advisory suggestions can time out under free-tier latency; the deterministic engine remains the primary authoritative gate, failing gracefully to labeled status without halting review.

5. RENDER FREE-TIER COLD STARTS

Backend services sleep after 15 min idle; initial container spin-up takes ~45 seconds. Handled gracefully with live 'waking' UI feedback and automatic retry cadence.

6. TRANSIENT INSIGHTS LATENCY

PostgreSQL newest row can be transiently stale under burst concurrent writes; refreshing or waiting a few seconds settles the aggregation counts.

7. BRAND-SWITCH UI SETTLEMENT

Switching brand workspaces can occasionally lag by one observation cycle before UI state fully settles to the active brand's cached version.

8. SUBMIT TITLE EDITS

Title-only edits in the Submit form do not persist to the Report heading (governance evaluates draft body text, attachments, and metadata).

9. TREND 'STABLE' CLASSIFICATION

The 'Stable' trend determination in Insights v2 is verified by local test fixtures, but has not been observed live during continuous production traffic (which yielded IMPROVING or WORSENING).

10. MOBILE 360PX & 120% ZOOM

Layout constraints and responsive classes are implemented, but physical mobile (360px) and 120% browser zoom were not verified in a physical browser harness.